

Discrete Probability Distributions

Flight Delay Analysis & Journal Article Review

STAT 562 Project 1

Introduction

Data: 3,000,000 flight-level records with FL_DATE and DEP_DELAY. Cleaning retained 2,922,356 flights with nonmissing date and departure delay.

Observed calendar: 1,704 daily intervals from 2019-01-01 to 2023-08-31.

Delayed flight: $\text{DEP_DELAY} > 0$ minutes.

Mean departure delay: 10.123 minutes. Empirical SD: 49.252 minutes.

Delayed flights: 992,843, or 33.974%.

Binomial Distribution

Use: high-delay days in a fixed 10-day period.

Own-data success: daily delayed-flight rate > 40%.

Estimated $p = 0.275235$; $E[X] = 2.752$; $\text{Var}(X) = 1.995$.

Simulation, 10,000 draws: mean = 2.756; variance = 1.989.

MGF: $M_X(t) = (1-p+pe^t)^n$, $t \in \mathbb{R}$.

Theory: Blitzstein and Hwang (2019).

Poisson Distribution

Own-data outcome: delayed flights per observed day.

$\lambda = 582.654$; empirical variance = 73,845.030; empirical SD = 271.744.

Poisson theoretical variance = 582.654; variance-to-mean ratio = 126.739.

Simulation mean = 582.747; simulation variance = 583.783.

MGF: $M_X(t) = \exp\{\lambda(e^t - 1)\}$, $t \in \mathbb{R}$.

Theory: Blitzstein and Hwang (2019).

Geometric Distribution

Own-data success: at least 10% of a day's flights have DEP_DELAY > 30 minutes.

$p = 0.487676$; $E[X] = 2.051$ days; $\text{Var}(X) = 2.154$.

Simulation mean = 2.061; simulation variance = 2.165.

MGF: $M_X(t) = p e^t / [1 - (1-p)e^t]$, $t < -\log(1-p)$.

Theory: Blitzstein and Hwang (2019).

Negative Binomial Distribution

Own-data success: a delayed flight, $\text{DEP_DELAY} > 0$.

$r = 5$; $p = 0.339741$; $E[X] = 14.717$ flights; $\text{Var}(X) = 28.602$.

Simulation mean = 14.748; simulation variance = 28.426.

MGF: $[p e^t / \{1-(1-p)e^t\}]^r$, $t < -\log(1-p)$.

Theory: Blitzstein and Hwang (2019).

Hypergeometric Distribution

Own-data population: $N = 2,922,356$; delayed flights $K = 992,843$; sample $n = 100$.

$E[X] = 33.974$; $\text{Var}(X) = 22.431$.

Simulation mean = 33.932; simulation variance = 22.539.

MGF: sum over support of $e^{tx} P(X=x)$, t in $\mathbb{R}$.

Theory: Blitzstein and Hwang (2019).

Journal Article: Aircraft Delay Analysis

Peer-reviewed journal article: Mitsokapas, Schäfer, Harris, and Beck (2021), Scientific Reports.

The study statistically characterized arrival delays at UK airports during 2018–2020 and reported broad delay behavior.

Connection: the project's daily count variance greatly exceeds its Poisson mean, so a simple Poisson model is a benchmark rather than a fitted conclusion.

Mueller and Chatterji (2002) remains background: it is an AIAA technical-forum paper, not the journal article used here.

Conclusion

The flight-level results support five distinct probability experiments.

Poisson provides a count benchmark but overdispersion is substantial.

Revised daily-rate thresholds avoid the earlier degenerate Binomial and Geometric results.

All five 10,000-draw simulations closely match their theoretical moments.

Interpret results as describing this sample, not every scheduled U.S. flight.

References

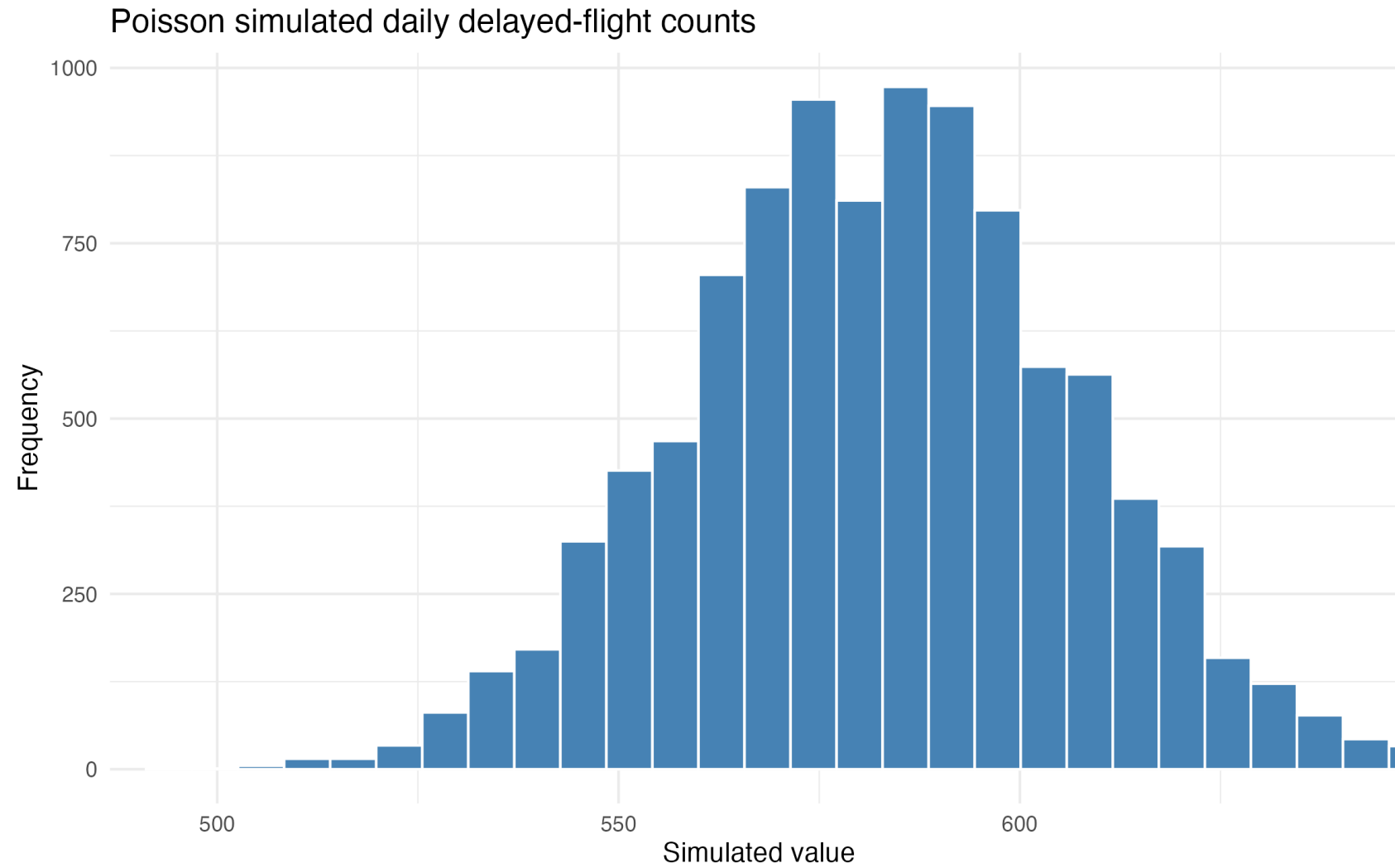
Blitzstein, J. K., & Hwang, J. (2019). Introduction to Probability (2nd ed.). Chapman and Hall/CRC.

Mitsokapas, E., Schäfer, B., Harris, R. J., & Beck, C. (2021). Statistical characterization of airplane delays. Scientific Reports, 11, 7855. <https://doi.org/10.1038/s41598-021-87279-8>

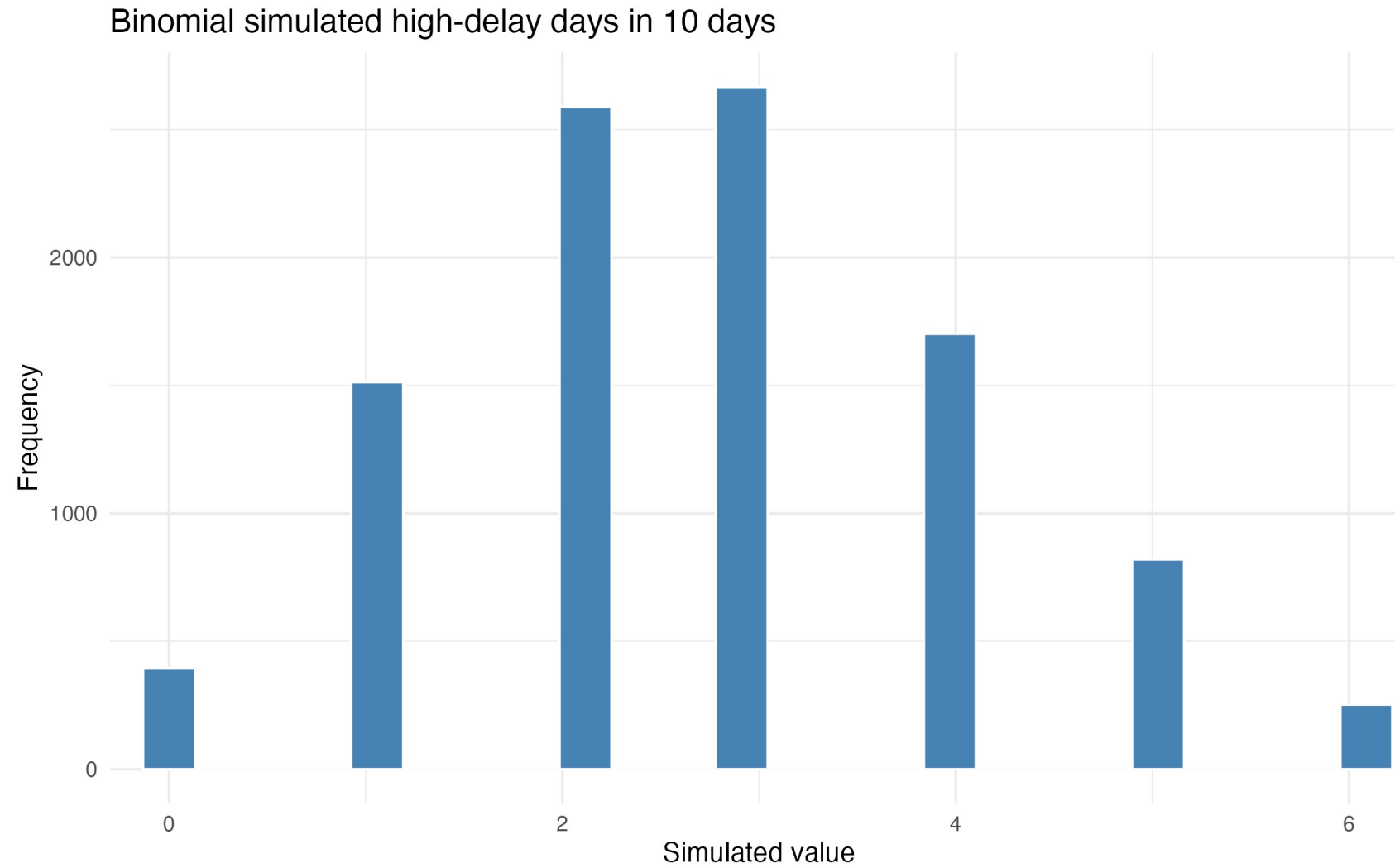
Mueller, E. R., & Chatterji, G. B. (2002). Analysis of aircraft arrival and departure delay characteristics. AIAA ATIO Technical Forum, AIAA 2002-5866. <https://doi.org/10.2514/6.2002-5866>

Flight-level data: flights_sample_3m.csv, source citation to be completed from original download.

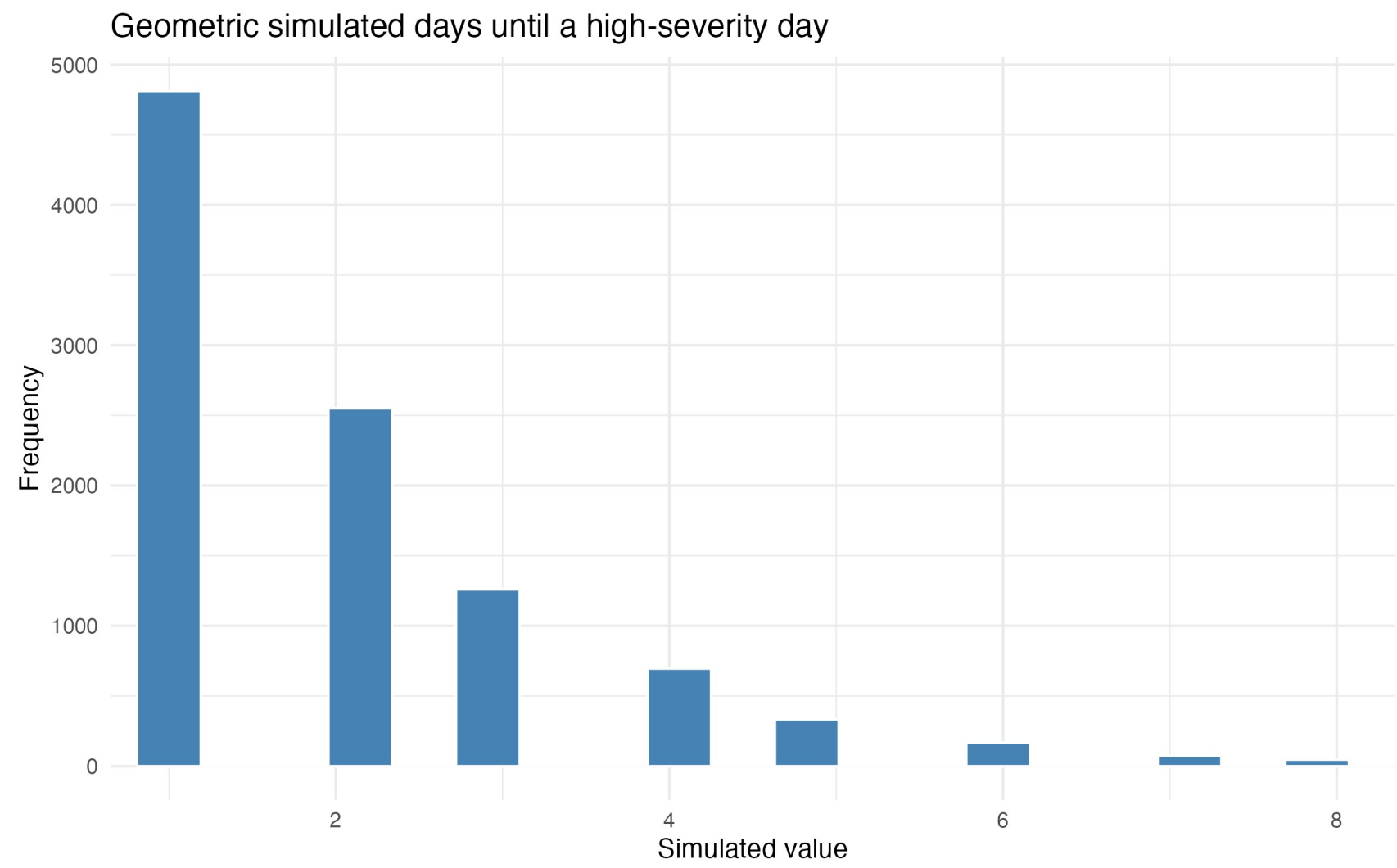
Poisson simulation



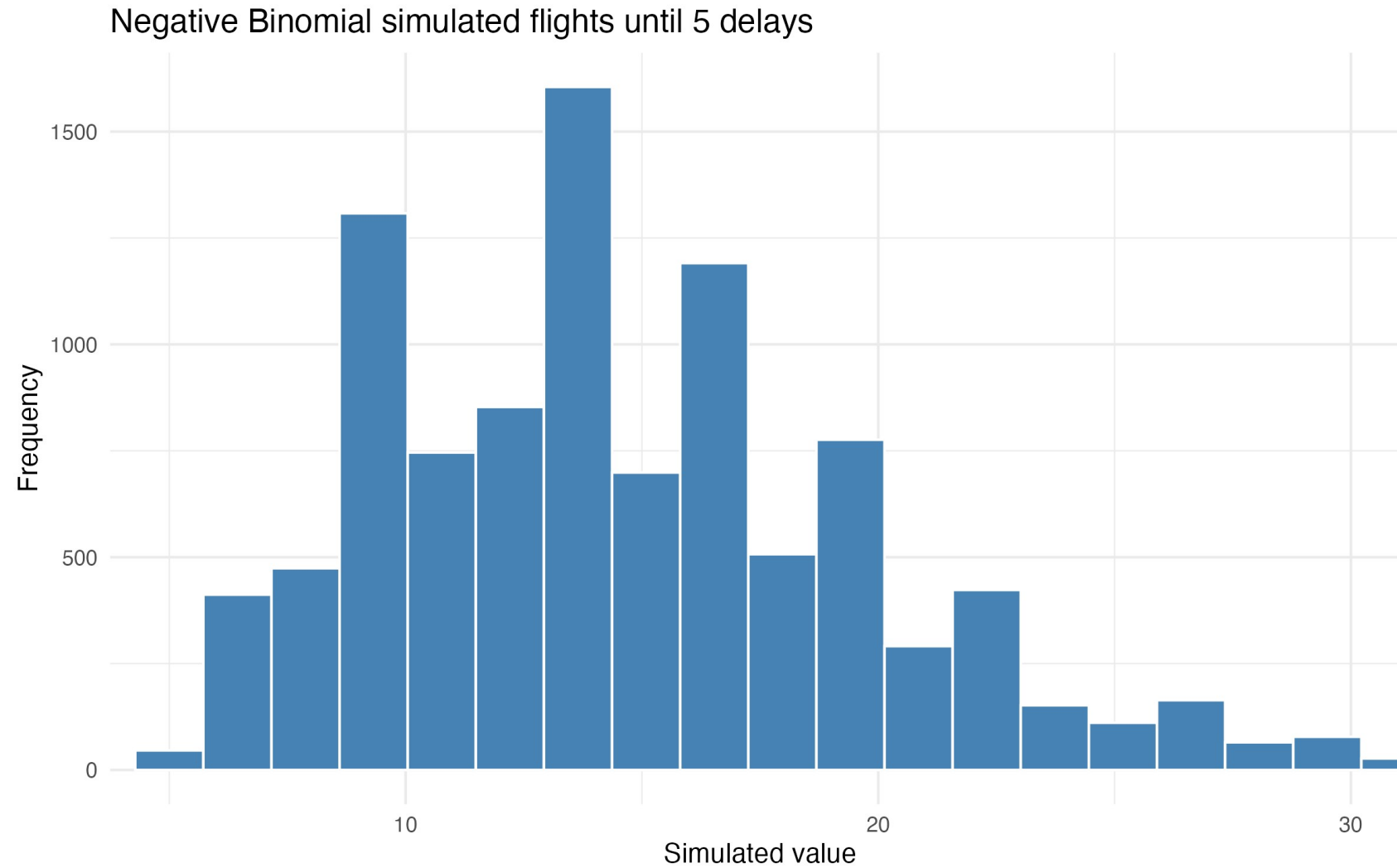
Binomial simulation



Geometric simulation



Negative Binomial simulation



Hypergeometric simulation

